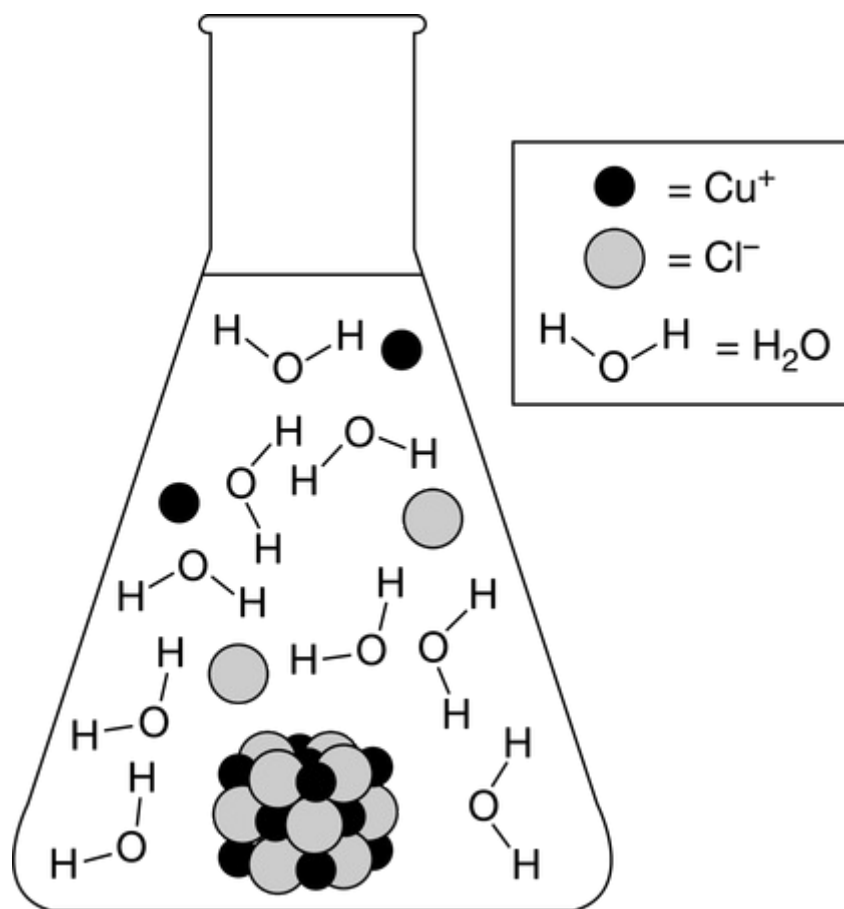
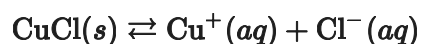


## Unit 7 Progress Check: MCQ



The particle diagram shown above represents the dissolution of  $\text{CuCl}(s)$  assuming an equilibrium concentration for  $\text{Cu}^+$  ions of about  $4 \times 10^{-4} M$  in a saturated solution at  $25^\circ\text{C}$ . The equilibrium being represented is shown in the following chemical equation.



1. Which of the following changes to the particle diagram will best represent the effect of adding  $1.0 \text{ mL}$  of  $4 M \text{ NaCl}$  to the solution?



**Unit 7 Progress Check: MCQ**

---

(A) Some of the  $\text{Cu}^+$  and  $\text{Cl}^-$  ions combine to form  $\text{CuCl}(s)$  because the  $K_{sp}$  will be lower than  $1.6 \times 10^{-7}$ .

(B) Some of the  $\text{Cu}^+$  and  $\text{Cl}^-$  ions combine to form  $\text{CuCl}(s)$  because the molar solubility will be lower than  $4 \times 10^{-4} M$ . ✓

(C) More  $\text{Cu}^+$  and  $\text{Cl}^-$  ions will be in solution because the molar solubility will be higher than  $4 \times 10^{-4} M$ .

(D) More  $\text{Cu}^+$  and  $\text{Cl}^-$  ions will be in solution because the  $K_{sp}$  will be higher than  $1.6 \times 10^{-7}$ .

2. Which of the following best explains what the particle diagram is able to show about the entropy change for the dissolution of  $\text{CuCl}(s)$  in pure water?

(A) The particle diagram shows that the dissociation of  $\text{CuCl}(s)$  into ions contributes to an increase in the entropy for the dissolution. ✓

(B) The particle diagram shows that the dissociation of  $\text{CuCl}(s)$  into ions contributes to a decrease in the entropy for the dissolution.

(C) The particle diagram shows that there is no reorganization of the water molecules around the ions and the change in entropy for the dissolution is zero.

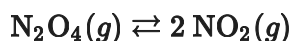
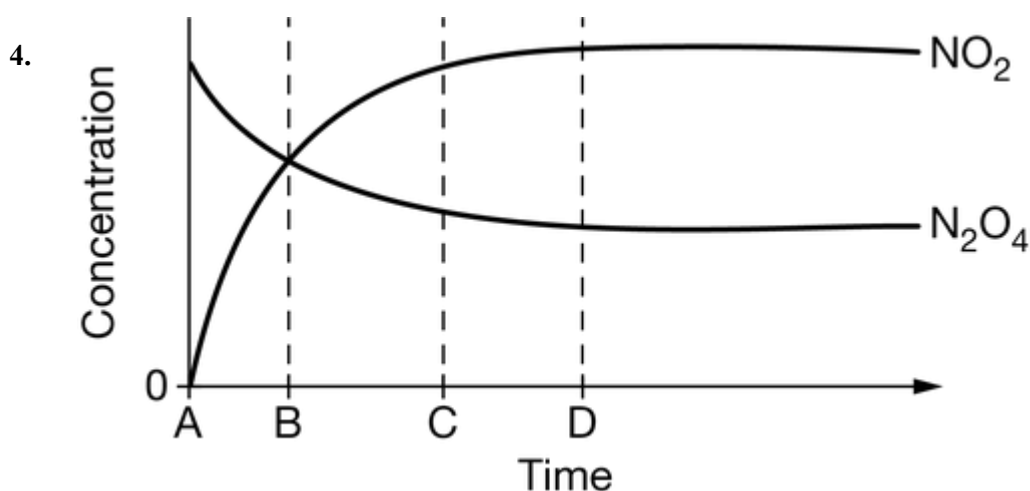
(D) The particle diagram shows that there are no interactions between the water molecules and the change in entropy for the dissolution is zero.

3. Which of the following best explains whether or not the particle diagram can predict the relative value of the enthalpy change for the dissolution of  $\text{CuCl}(s)$  ?



## Unit 7 Progress Check: MCQ

- The value of the enthalpy change for the dissolution of  $\text{CuCl}(s)$  cannot be predicted from the particle diagram because it fails to illustrate the amount of energy required to overcome the forces between solute particles and between solvent particles. ✓
- The value of the enthalpy change for the dissolution of  $\text{CuCl}(s)$  cannot be predicted from the particle diagram because it fails to illustrate the amount of energy released when the water molecules form hydrogen bonds with  $\text{Cl}^-$  ions.
- The value of the enthalpy change for the dissolution of  $\text{CuCl}(s)$  is positive (endothermic) because energy is released to overcome the forces between solute particles, as shown in the particle diagram.
- The value of the enthalpy change for the dissolution of  $\text{CuCl}(s)$  is negative (exothermic) because energy is required when the bonds between the ions are broken, as shown in the particle diagram.



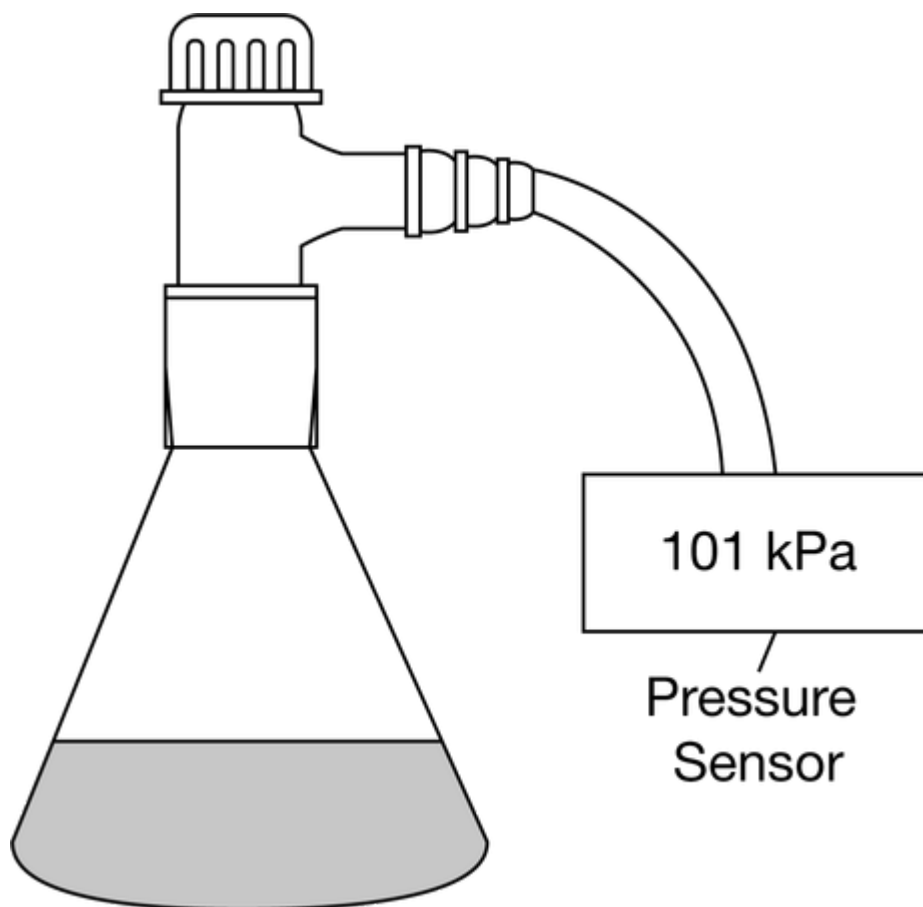
The graph above represents the data collected under certain conditions for the decomposition of  $\text{N}_2\text{O}_4(g)$  according to the chemical equation above. Based on the graph, at approximately which time is equilibrium established?

- At time A, because  $\text{N}_2\text{O}_4(g)$  is expanding to fill the container.
- At time B, because the reaction is reversible and  $[\text{NO}_2] = [\text{N}_2\text{O}_4]$ .
- At time C, because the reaction is about to reach completion and  $[\text{NO}_2] > [\text{N}_2\text{O}_4]$ .
- At time D, because there are no observable changes in  $[\text{NO}_2]$  and  $[\text{N}_2\text{O}_4]$ . ✓



## Unit 7 Progress Check: MCQ

5.



| Time (min) | Pressure (kPa) |
|------------|----------------|
| 0.25       | 101            |
| 0.50       | 108            |
| 0.75       | 116            |
| 1.00       | 124            |
| 1.25       | 128            |
| 1.50       | 130            |
| 1.75       | 130            |
| 2.00       | 130            |

A sample of acetone is placed into a container. The container is sealed and attached to a pressure sensor, as shown in the diagram above. The container is allowed to sit on the lab table for a few minutes as the pressure in the container is monitored at regular intervals. At the end of 2.00 minutes, some acetone liquid remains in the container. Which of the following best explains the pressure data presented in the table above?

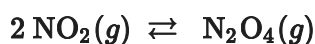


**Unit 7 Progress Check: MCQ**

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- (A) The acetone heats up over time, causing more of it to vaporize at an increasing rate.
- (B) The acetone has completely vaporized after 1.50 minutes, so the pressure becomes constant.
- (C) The acetone vaporizes from the liquid at a constant rate and the rate of condensation increases until it becomes equal to the rate of evaporation and the pressure stays constant. ✓
- (D) The acetone vaporizes from the liquid at a rate that is fast in the beginning but then slows down until the vaporization process stops completely.

6.      **Brown**      **Colorless**



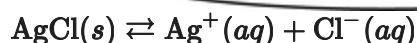
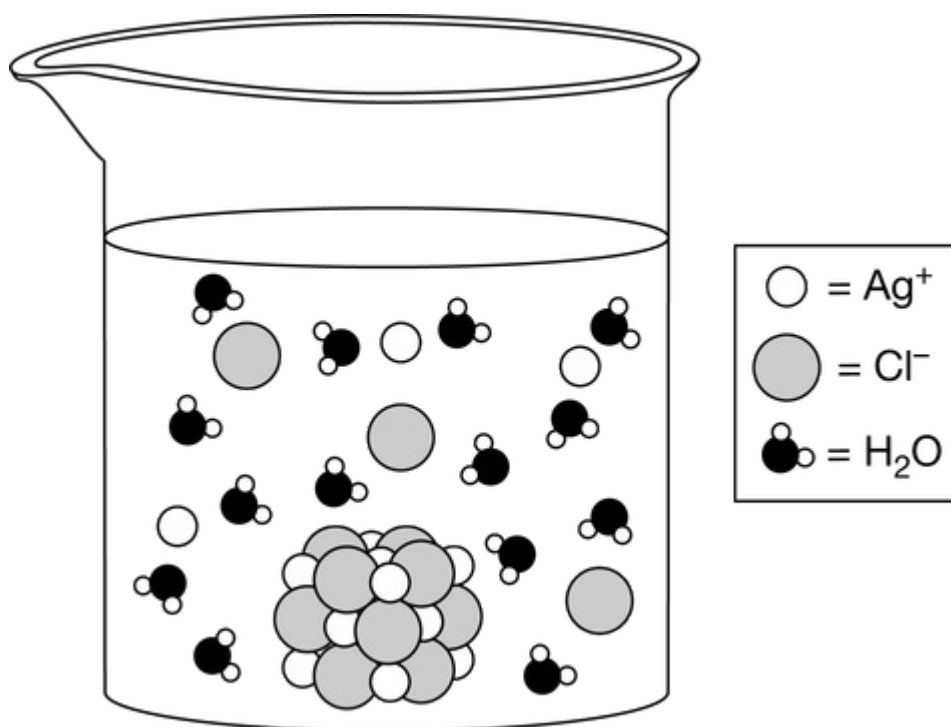
A sample of pure  $\text{NO}_2(g)$  in a sealed tube at  $20^\circ\text{C}$  is placed in a temperature bath at  $30^\circ\text{C}$ . Observations of changes in the color, pressure, and mass of the mixture are recorded as a function of time. Which of the following is an observation that would best support the claim that the reaction represented above has reached equilibrium at  $30^\circ\text{C}$ ?

- (A) The total mass of the system remains constant because at equilibrium the amounts of reactant and product do not change with time.
- (B) The color of the system changes from brown to completely colorless because at equilibrium only the product will be present.
- (C) The total pressure of the system decreases then reaches a constant value because at equilibrium the amounts of reactant and product no longer change. ✓
- (D) The temperature of the system remains constant because at equilibrium the temperature must be constant.



## Unit 7 Progress Check: MCQ

7.



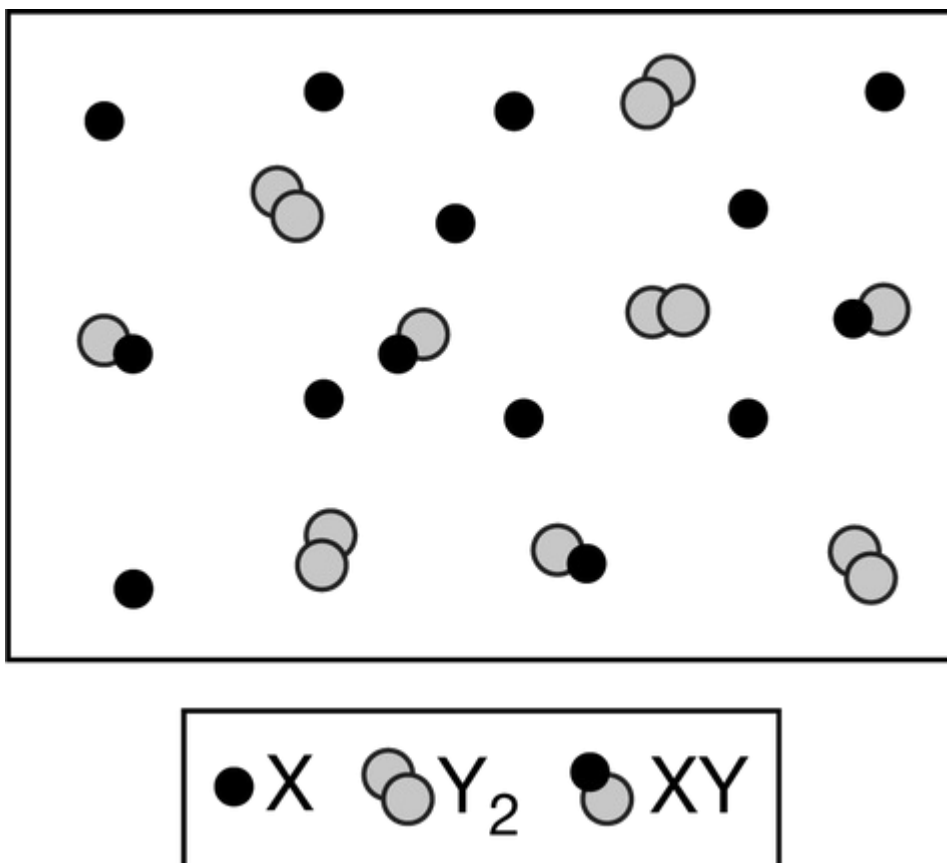
When  $\text{AgCl}(s)$  is placed in water, it dissolves according to the chemical equation above. The particle diagram above was proposed to represent an aqueous solution in which  $\text{AgCl}(s)$  is in equilibrium with its ions. Which of the following best explains whether or not the diagram provides a good representation of this dynamic equilibrium at the microscopic level?

- (A) It is a good representation because it shows that the dipoles of the  $\text{H}_2\text{O}$  molecules are oriented around the  $\text{Ag}^+$  and  $\text{Cl}^-$  ions in solution.
- (B) It is a good representation because it shows that the concentrations of  $\text{Ag}^+$  ions and  $\text{Cl}^-$  ions are equal at equilibrium.
- (C) It is not a good representation because it does not show that the concentration of  $\text{AgCl}(s)$  is constant at equilibrium.
- (D) It is not a good representation because it does not illustrate the dynamic equilibrium in which the rates of the forward and reverse reactions are equal. ✓



## Unit 7 Progress Check: MCQ

8.



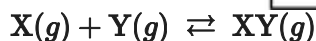
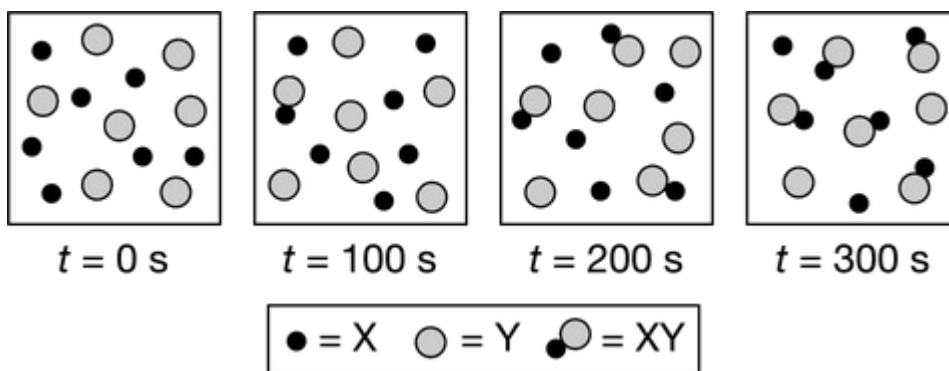
The particle diagram above shows the system represented by the equation  $2 \text{X}(g) + \text{Y}_2(g) \rightleftharpoons 2 \text{XY}(g)$ . Which of the following explains whether the particle diagram indicates that the system is at equilibrium?

- (A) The particle diagram does not indicate that the system is at equilibrium because it shows the system only at one point in time. ✓
- (B) The particle diagram does not indicate that the system is at equilibrium because the ratio  $\frac{[\text{XY}]_{eq}}{[\text{X}]_{eq}}$  is not equal to 1.
- (C) The particle diagram indicates that the system is at equilibrium because the value of  $K$  is small.
- (D) The particle diagram indicates that the system is at equilibrium because  $[\text{X}]_{eq} = 2 \times [\text{Y}_2]_{eq}$ .



## Unit 7 Progress Check: MCQ

9.



The particle diagrams above show the changes that occurred after an equimolecular mixture of  $\text{X}(g)$  and  $\text{Y}(g)$  was placed inside a rigid container at constant temperature. Which of the following statements is best supported by the particle diagrams?

- (A) The rate of the reverse reaction is zero because the concentration of  $\text{XY}$  keeps increasing.
- (B) The reverse reaction has a higher rate than the forward reaction between **200 s** and **300 s** because  $[\text{XY}] \gg [\text{X}]$ .
- (C) The forward reaction has a faster rate than the reverse reaction between **0 s** and **300 s** because more products were being formed. ✓
- (D) The rates of the forward and reverse reactions were the same after **100 s** because the rate of formation of  $\text{XY}$  was constant.

10.  $2 \text{HI}(g) \rightleftharpoons \text{H}_2(g) + \text{I}_2(g)$ 

| Substance            | Initial Concentration ( $M$ ) | Equilibrium Concentration ( $M$ ) |
|----------------------|-------------------------------|-----------------------------------|
| <b>HI</b>            | 0.30                          | 0.24                              |
| <b>H<sub>2</sub></b> | 0.00                          | 0.030                             |
| <b>I<sub>2</sub></b> | 0.00                          | ?                                 |

In an experiment involving the reaction shown above, a sample of pure **HI** was placed inside a rigid container at a certain temperature. The table above provides the initial and equilibrium concentrations for some of the substances in the reaction. Based on the data, which of the following is the value of the equilibrium constant ( $K_{eq}$ ) for the reaction, and why?



## Unit 7 Progress Check: MCQ

(A)  $K_{eq} = 2.5 \times 10^{-1}$ , because  $[I_2]_{eq} = 2 \times [HI]_{eq}$ .

(B)  $K_{eq} = 6.3 \times 10^{-2}$ , because  $[I_2]_{eq} = \frac{1}{2}[HI]_{eq}$ .

(C)  $K_{eq} = 1.6 \times 10^{-2}$ , because  $[I_2]_{eq} = [H_2]_{eq}$ . ✓

(D)  $K_{eq} = 3.1 \times 10^{-2}$ , because  $[I_2]_{eq} = 2 \times [H_2]_{eq}$ .

11.  $SO_2(g) + Cl_2(g) \rightleftharpoons SO_2Cl_2(g)$   
 $SO_2(g)$  reacts with  $Cl_2(g)$  to produce  $SO_2Cl_2(g)$ , according to the equation above. A mixture of **1.00 atm** of  $SO_2(g)$  and **1.00 atm** of  $Cl_2(g)$  is placed in a rigid, evacuated **1.00 L** container, and the reaction mixture is allowed to reach equilibrium at a certain temperature. The table below shows the initial partial pressure of each gas and the equilibrium partial pressures of  $SO_2(g)$  and  $SO_2Cl_2(g)$  at the same temperature.

|                            | $P_{SO_2}$ | $P_{Cl_2}$ | $P_{SO_2Cl_2}$ |
|----------------------------|------------|------------|----------------|
| Initial pressure (atm)     | 1.00       | 1.00       | 0.00           |
| Equilibrium pressure (atm) | 0.20       | ?          | 0.80           |

Which of the following indicates the equilibrium value of  $P_{Cl_2}$  and the approximate value of the equilibrium constant  $K_p$ ?

(A)  $P_{Cl_2} = 0.10 \text{ atm}$  and  $K_p = 5.0$

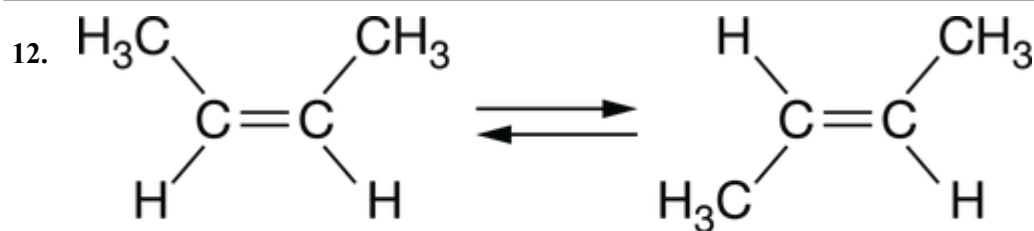
(B)  $P_{Cl_2} = 0.20 \text{ atm}$  and  $K_p = 0.050$

(C)  $P_{Cl_2} = 0.20 \text{ atm}$  and  $K_p = 20$ . ✓

(D)  $P_{Cl_2} = 0.80 \text{ atm}$  and  $K_p = 0.025$

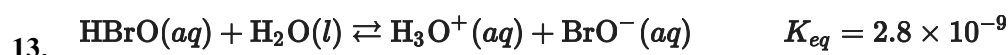


## Unit 7 Progress Check: MCQ

*Cis*-2-Butene*Trans*-2-Butene

The diagram above represents the equilibrium between two isomers of 2-butene. The equilibrium constant,  $K_c$ , is 1.2 at a certain temperature. Two identical vessels each contain an equilibrium mixture of the two gases at that temperature. The concentration of *cis*-2-butene in the second vessel is twice the concentration in the first vessel. What is the concentration of *trans*-2-butene in the second vessel compared to that in the first vessel?

- (A) Half the concentration of that in the first vessel
- (B) The same concentration as that in the first vessel
- (C) Twice the concentration of that in the first vessel ✓
- (D) Four times the concentration of that in the first vessel

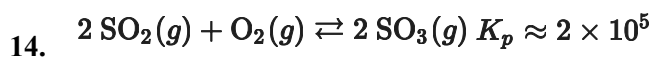


The equilibrium reaction in  $0.100\text{ M HBrO}(aq)$  at equilibrium is represented by the equation above. Based on the magnitude of the equilibrium constant, which of the following correctly compares the equilibrium concentrations of substances involved in the reaction, and why?

- (A) The equilibrium concentration of  $\text{BrO}^-$  will be much smaller than the equilibrium concentration of  $\text{H}_3\text{O}^+$ , because  $\text{H}_2\text{O}$  is the solvent and is present in the largest amount.
- (B) The equilibrium concentration of  $\text{BrO}^-$  will be much smaller than the equilibrium concentration of  $\text{HBrO}$ , because  $K_{eq} \ll 1$ . ✓
- (C) The equilibrium concentration of  $\text{H}_3\text{O}^+$  will be much smaller than the equilibrium concentration of  $\text{BrO}^-$ , because all the  $\text{HBrO}$  will react to produce  $\text{BrO}^-$ .
- (D) The equilibrium concentration of  $\text{H}_3\text{O}^+$  will be much larger than the equilibrium concentration of  $\text{HBrO}$ , because  $K_{eq} \ll 1$ .



## Unit 7 Progress Check: MCQ



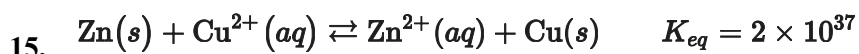
At a certain temperature,  $\text{SO}_2(g)$  and  $\text{O}_2(g)$  react to produce  $\text{SO}_3(g)$  according to the chemical equation shown above. An evacuated rigid vessel is originally filled with  $\text{SO}_2(g)$  and  $\text{O}_2(g)$ , each with a partial pressure of **1 atm**. Which of the following is closest to the partial pressure of  $\text{O}_2(g)$  after the system has reached equilibrium, and why?

(A) **0 atm**; because  $K_p$  is very large, nearly all the  $\text{SO}_2(g)$  and  $\text{O}_2(g)$  are consumed before the system reaches equilibrium.

(B) **0.5 atm**; because  $K_p$  is very large, nearly all the  $\text{SO}_2(g)$  is consumed before the system reaches equilibrium, but an excess amount of  $\text{O}_2(g)$  remains at equilibrium. ✓

(C) **1 atm**; because  $K_p$  is very large, the system is already near equilibrium, and there will be very little change to the partial pressure of  $\text{O}_2(g)$ .

(D) **1.5 atm**; because  $K_p$  is very large, the decomposition of any  $\text{SO}_3(g)$  that forms increases the amount of  $\text{O}_2(g)$  at equilibrium.



A **0.10 mol** sample of solid zinc was added to **500.0 mL** of **1.0 M**  $\text{Cu}(\text{NO}_3)_2(aq)$ . After the mixture sits overnight, which of the following best describes what was most likely observed and measured the next morning and why?

(A) Almost all of the  $\text{Zn}(s)$  will still be in the beaker with no visible  $\text{Cu}(s)$ , because equilibrium was not reached due to the very large  $K_{eq}$ .

(B) About half of the  $\text{Zn}(s)$  will have disappeared and  $\text{Cu}(s)$  will have appeared in the beaker, because the system reached equilibrium.

(C) About two-thirds of the  $\text{Zn}(s)$  had disappeared and  $\text{Cu}(s)$  will have appeared in the beaker, because the system reached equilibrium.

(D) Virtually all of the  $\text{Zn}(s)$  will have disappeared and  $\text{Cu}(s)$  will have appeared in the beaker, because the reaction went almost to completion at equilibrium due to the very large  $K_{eq}$ . ✓



## Unit 7 Progress Check: MCQ

| 16. | Reaction | Chemical Equation                                                            | $K_{eq}$         |
|-----|----------|------------------------------------------------------------------------------|------------------|
|     | 1        | $\text{HA}(aq) \rightleftharpoons \text{H}^+(aq) + \text{A}^-(aq)$           | $K_1 = 0.010$    |
|     | 2        | $\text{H}_2\text{B}(aq) \rightleftharpoons \text{H}^+(aq) + \text{HB}^-(aq)$ | $K_2 = 0.000010$ |

The table above provides the  $K_{eq}$  values for two reactions. Which of the following is the correct mathematical expression needed to determine the equilibrium constant of the reaction shown below?

Reaction 3:  $\text{HA}(aq) + \text{HB}^-(aq) \rightleftharpoons \text{H}_2\text{B}(aq) + \text{A}^-(aq)$   $K_{eq} = K_3$

(A)  $K_3 = K_1 \times \frac{1}{K_2}$  ✓

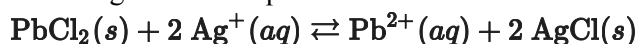
(B)  $K_3 = \frac{1}{K_1} \times K_2$

(C)  $K_3 = K_1 - K_2$

(D)  $K_3 = -K_1 + K_2$

|     |             |                                                                              |                |
|-----|-------------|------------------------------------------------------------------------------|----------------|
| 17. | Reaction 1: | $\text{PbCl}_2(s) \rightleftharpoons \text{Pb}^{2+}(aq) + 2 \text{Cl}^-(aq)$ | $K_{sp} = K_1$ |
|     | Reaction 2: | $\text{AgCl}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{Cl}^-(aq)$        | $K_{sp} = K_2$ |

Based on the information given above, which of the following is the expression for  $K_{eq}$  for the reaction that occurs when a  $0.1 \text{ M AgNO}_3(aq)$  is added to a saturated solution of  $\text{PbCl}_2(aq)$ , as represented by the following chemical equation?



(A)  $K_{eq} = K_1 + (2 \times K_2)$

(B)  $K_{eq} = K_1 - (2 \times K_2)$

(C)  $K_{eq} = K_1 \times (K_2)^2$

(D)  $K_{eq} = \frac{K_1}{(K_2)^2}$  ✓



## Unit 7 Progress Check: MCQ

18.

|             |                                                               |       |
|-------------|---------------------------------------------------------------|-------|
| Equation 1: | $2 \text{NO}_2(g) \rightleftharpoons \text{N}_2\text{O}_4(g)$ | $K_1$ |
| Equation 2: | $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$ | $K_2$ |

In a large reaction vessel at a constant temperature, nitrogen dioxide and dinitrogen tetroxide are in a state of dynamic equilibrium, as represented by the chemical equations shown above. The equilibrium constants for the reactions are  $K_1$  and  $K_2$ . Which of the following quantities can most easily be used to find the value of  $K_2$ ?

- (A) The value of  $\Delta H$  for the reaction
- (B) The temperature of the system
- (C) The volume of the system

(D) The value of  $K_1$



19.  $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq}) + 4\text{Cl}^{-}(\text{aq}) \rightleftharpoons \text{CoCl}_4^{2-}(\text{aq}) + 6\text{H}_2\text{O}(\text{l}) \quad \Delta H > 0$   
pink blue

A student poured **10 mL** of **CoCl<sub>2</sub>(aq)** into a test tube and added a few drops of concentrated **HCl**, which resulted in a deep-blue solution. The reaction that occurred is represented by the chemical equation shown above. Then, the student placed the test tube inside a beaker that contained ice and water for about five minutes. Which of the following describes what the student most likely observed next, and why?

- (A) The color of the solution changed from blue to pink, because lowering the temperature increased the collision frequency between  $\text{CoCl}_4^{2-}$  and  $\text{H}_2\text{O}$ .

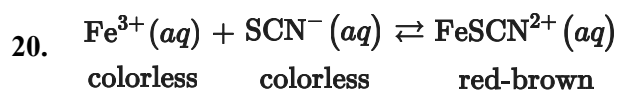
(B) The color of the solution changed from blue to pink, because cooling caused the equilibrium to shift to form the pink-colored  $\text{Co}(\text{H}_2\text{O})_6^{2+}$ .



- (C)** The color of the solution did not change, because more water was not added to the solution.
- (D)** The color of the solution did not change, because heat is not released from this reaction.



## Unit 7 Progress Check: MCQ

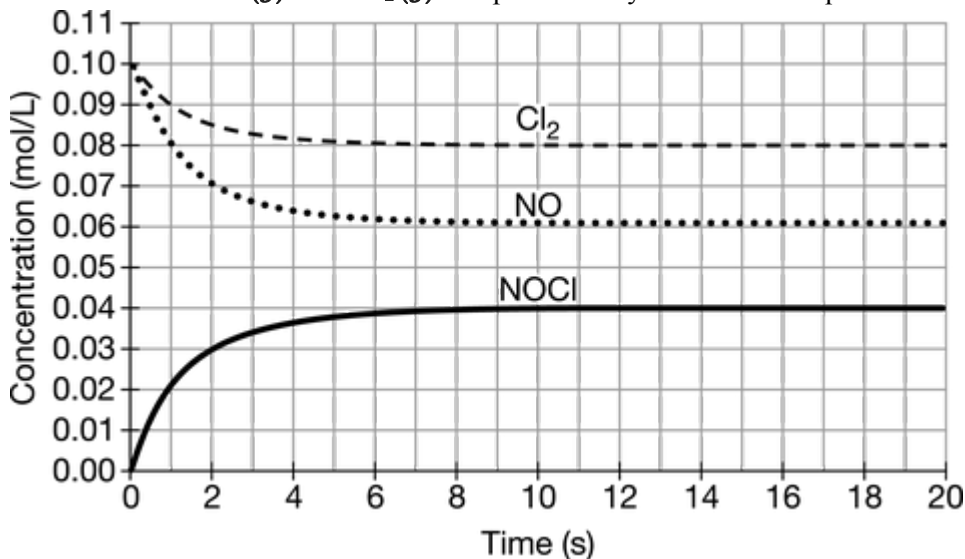


The formation of  $\text{FeSCN}^{2+}$  in an aqueous solution is represented by the chemical equilibrium shown above. A light red-brown solution is prepared combining **12.50 mL** of **0.5 M  $\text{Fe}(\text{NO}_3)_3$** , **0.5 mL** of **0.002 M KSCN**, and **37.0 mL** of water that had been slightly acidified. If an additional **1.0 mL** of **0.5 M  $\text{Fe}(\text{NO}_3)_3$**  is added, which of the following predicts and explains correctly whether the darkness of the color of the solution will change?

- (A) The color of the solution will lighten because most of the volume added is solvent.
- (B) The color of the solution will darken because the equilibrium will favor the dissociation of  $\text{FeSCN}^{2+}$ .
- (C) The color of the solution will darken because the equilibrium will favor the formation of more  $\text{FeSCN}^{2+}$ . ✓
- (D) The color of the solution will not change because the solution already contains  $\text{Fe}^{3+}$  ions.



The reaction of  $\text{NO}(\text{g})$  with  $\text{Cl}_2(\text{g})$  is represented by the balanced equation above.

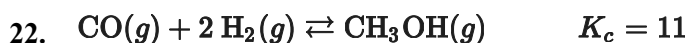


A chemist carried out the reaction at **573 K**, starting with **0.100 mol** of each reactant in a **1.00 L** container with variable volume. The reaction mixture quickly reached equilibrium, as indicated in the graph above. The chemist reduces the volume of the equilibrium system by half while keeping the temperature constant. Which of the following predictions about the yield of the reaction is best, and why?



## Unit 7 Progress Check: MCQ

- By halving the volume, the pressure doubles. The system will respond to the increase in pressure by decreasing the total number of moles of gas in the system. Thus, the yield will increase because the reaction will shift toward more product. ✓
- By halving the volume, the pressure doubles. The system will respond to the increase in pressure by increasing the total number of moles of gas in the system. Thus, the yield will decrease because the reaction will shift toward more reactants.
- By halving the volume, the pressure is reduced by half. The system will respond to the decrease in pressure by increasing the total number of moles of gas in the system. Thus, the yield will decrease because the reaction will shift toward more reactants.
- By halving the volume, the pressure is reduced by half. The system will respond to the decrease in pressure by decreasing the total number of moles of gas in the system. Thus, the yield will increase because the reaction will shift toward more products.



| Substance          | Initial Concentration (M) |
|--------------------|---------------------------|
| CO                 | 0.10                      |
| H <sub>2</sub>     | 0.10                      |
| CH <sub>3</sub> OH | 0.10                      |

The table lists the initial concentrations of each substance in the system represented by the equation above at a given temperature. Which of the following best predicts what will occur as the system approaches equilibrium?

- The rate of the reverse reaction will be less than the rate of the forward reaction and additional  $\text{CH}_3\text{OH}(g)$  will be consumed, because  $K_c < Q_c$ .
- The rate of the reverse reaction will be greater than the rate of the forward reaction and additional  $\text{CH}_3\text{OH}(g)$  will be consumed because  $K_c < Q_c$ . ✓
- The rate of the forward reaction will be less than the rate of the reverse reaction and additional  $\text{CH}_3\text{OH}(g)$  will be produced, because  $K_c < Q_c$ .
- The rate of the forward reaction will be greater than the rate of the reverse reaction and additional  $\text{CH}_3\text{OH}(g)$  will be produced, because  $K_c < Q_c$ .



## Unit 7 Progress Check: MCQ

23.

| Substance | Concentration (M) |
|-----------|-------------------|
| $X_2$     | 0.10              |
| $Y_2$     | 0.30              |
| $XY$      | 0.20              |

$X_2(g) + Y_2(g) \rightleftharpoons 2 XY(g)$   $K_c = 25$  The chemical reaction shown above took place inside a rigid container at constant temperature. The table provides the concentrations of reactants and products at some point during the reaction. Based on this information, which of the following explains whether or not the reaction has reached equilibrium, and why?

- (A) The reaction is not at equilibrium because  $Q_c > K_c$ ; the forward reaction is favored in order to form more  $XY$ .
- (B) The reaction is not at equilibrium because  $Q_c < K_c$ ; the reverse reaction is favored in order to form more  $X_2$  and  $Y_2$ .
- (C) The reaction is not at equilibrium because  $Q_c < K_c$ ; the forward reaction is favored in order to form more  $XY$ . ✓
- (D) The reaction is at equilibrium because  $Q_c = K_c$  and more  $XY$  will not be formed.

24.  $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$

At a given temperature, the system represented by the chemical equation above is at equilibrium inside a rigid container. Which of the following explains how the system will restore equilibrium, based on the correct relationship between  $Q$  and  $K$ , after a certain amount of  $Cl_2(g)$  is added?

- (A) Since  $Q = \frac{[Cl_2][CO]}{[COCl_2]}$ , when  $Cl_2(g)$  is added to the system  $Q > K$  and the system will restore equilibrium by producing more  $COCl_2(g)$ . ✓
- (B) Since  $Q = \frac{[Cl_2][CO]}{[COCl_2]}$ , when  $Cl_2(g)$  is added to the system  $Q < K$  and the system will restore equilibrium by producing more  $CO(g)$ .
- (C) Since  $Q = \frac{[COCl_2]}{[CO][Cl_2]}$ , when  $Cl_2(g)$  is added to the system  $Q > K$  and the system will restore equilibrium by producing more  $COCl_2(g)$ .
- (D) Since  $Q = \frac{[COCl_2]}{[CO][Cl_2]}$ , when  $Cl_2(g)$  is added to the system  $Q > K$  and the system will restore equilibrium by producing more  $CO(g)$ .



## Unit 7 Progress Check: MCQ

25. For which of the following salts would the relationship between molar solubility,  $s$ , in  $\text{mol/L}$ , and the value of  $K_{sp}$  be represented by the equation  $K_{sp} = 4s^3$ ?

(A)  $\text{PbCO}_3$

(B)  $\text{Mg}_3(\text{PO}_4)_2$

(C)  $\text{Ag}_2\text{SO}_4$



(D)  $\text{MnS}$

26.  $\text{Hg}_2\text{I}_2(s) \rightleftharpoons \text{Hg}_2^{2+}(aq) + 2\text{I}^-(aq)$        $K_{sp} = [\text{Hg}_2^{2+}][\text{I}^-]^2$

A saturated solution of  $\text{Hg}_2\text{I}_2$  is at equilibrium at  $25^\circ\text{C}$  as represented by the equation above. If  $[\text{I}^-] = 4.6 \times 10^{-10} \text{ M}$  at equilibrium, which of the following gives the correct molar solubility,  $S$ , and  $K_{sp}$  for  $\text{Hg}_2\text{I}_2$ ?

(A)  $S = 4.6 \times 10^{-10} \text{ M}$ ;  $K_{sp} = (2.3 \times 10^{-10})(4.6 \times 10^{-10})^2$

(B)  $S = 4.6 \times 10^{-10} \text{ M}$ ;  $K_{sp} = (4.6 \times 10^{-10})(9.2 \times 10^{-10})^2$

(C)  $S = 2.3 \times 10^{-10} \text{ M}$ ;  $K_{sp} = (2.3 \times 10^{-10})(4.6 \times 10^{-10})^2$



(D)  $S = 2.3 \times 10^{-10} \text{ M}$ ;  $K_{sp} = (4.6 \times 10^{-10})(9.2 \times 10^{-10})^2$

27.  $\text{AgI}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{I}^-(aq)$        $K_{sp} = 8.3 \times 10^{-17}$  at 298 K

The dissolution of  $\text{AgI}$  is represented above. Which of the following shows the mathematical relationship between the molar solubility,  $S$ , of  $\text{AgI}$  and the  $K_{sp}$  at 298 K?



## Unit 7 Progress Check: MCQ

(A)  $S = 8.3 \times 10^{-17} \text{ mol/L}$

(B)  $S = \left( \frac{8.3 \times 10^{-17}}{2} \right) \text{ mol/L}$

(C)  $S = \sqrt{8.3 \times 10^{-17}} \text{ mol/L}$  ✓

(D)  $S = 2\sqrt{8.3 \times 10^{-17}} \text{ mol/L}$



The equilibrium in a saturated solution of  $\text{Ca(OH)}_2$  is represented above. In an experiment, a student places 5.0 g of  $\text{Ca(OH)}_2(s)$  into 100.0 mL of distilled water and stirs the mixture. How would the results be affected if the student repeats the experiment but this time places 5.0 g of  $\text{Ca(OH)}_2(s)$  into 100.0 mL of 0.0010 M  $\text{NaOH}(aq)$  instead of distilled water?

(A) Less solid will dissolve, because the larger value of  $[\text{OH}^-]$  will cause the equilibrium position to lie farther to the right.

(B) Less solid will dissolve, because the larger value of  $[\text{OH}^-]$  will cause the equilibrium position to lie farther to the left. ✓

(C) More solid will dissolve, because the larger value of  $[\text{OH}^-]$  will cause the equilibrium position to lie farther to the right.

(D) More solid will dissolve, because the smaller value of  $[\text{OH}^-]$  will cause the equilibrium position to lie farther to the left.



The dissolution of solid  $\text{AgCN}$  is represented by the chemical equation above. In pure water, the equilibrium concentration of  $\text{Ag}^+$  ions in a saturated solution is  $7.7 \times 10^{-9} \text{ M}$ . If a small amount of solid  $\text{NaCN}$  is added to the saturated  $\text{AgCN}$  solution, which of the following would be observed?



## Unit 7 Progress Check: MCQ

- (A) The  $K_{sp}$  increases and more  $\text{AgCN}$  dissolves.
- (B) The  $K_{sp}$  increases and some  $\text{AgCN}$  precipitates.
- (C) The molar solubility of  $\text{AgCN}$  becomes smaller than  $7.7 \times 10^{-9} \text{ M}$  and some  $\text{AgCN}$  precipitates. ✓
- (D) The molar solubility of  $\text{AgCN}$  becomes larger than  $7.7 \times 10^{-9} \text{ M}$  and more  $\text{AgCN}$  dissolves.

30.

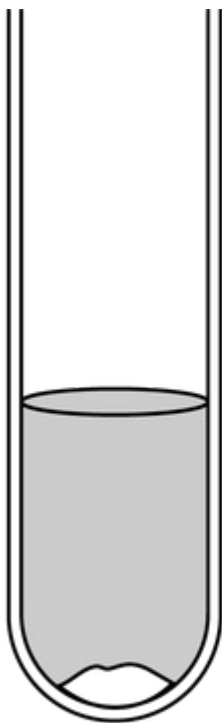


Figure 1

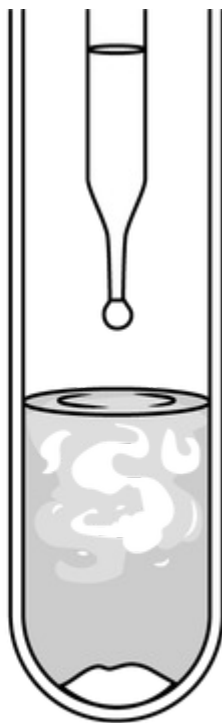
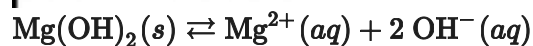
 $\text{pH} = 10.35$ 

Figure 2

A student prepared a saturated aqueous solution of  $\text{Mg}(\text{OH})_2$  and measured its  $\text{pH}$ , as shown in Figure 1 above. Then the student added a few drops of an unknown solution to the test tube and observed cloudiness in the solutions as shown in Figure 2. On the basis of this information and the equilibrium represented above, which of the following is most likely the identity of the reagent added from the dropper?



## Unit 7 Progress Check: MCQ

(A) Distilled water

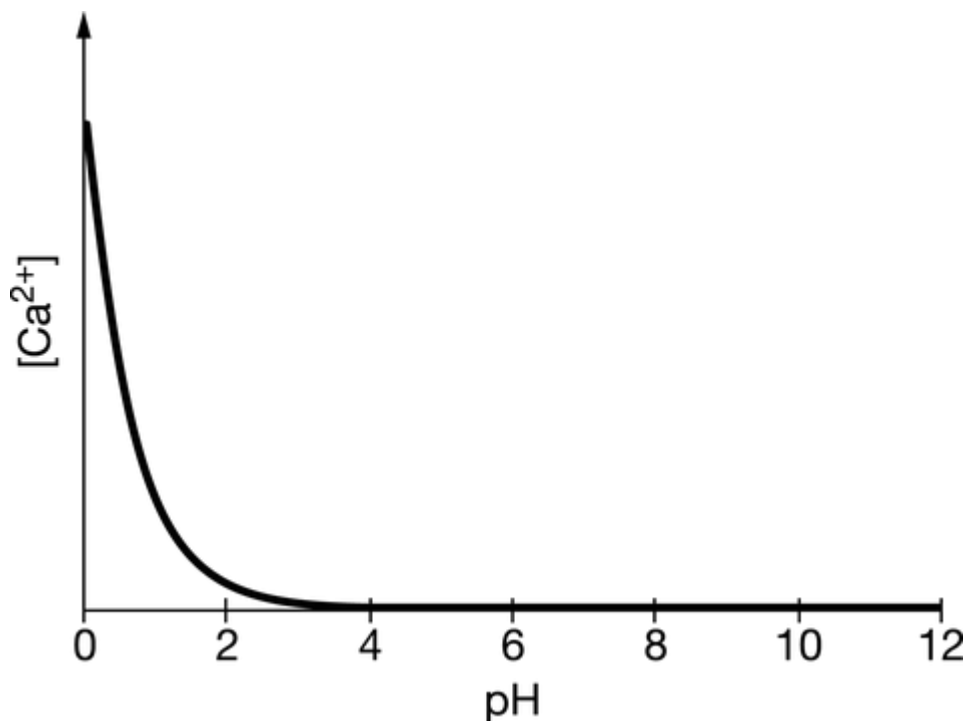
(B)  $\text{NaNO}_3(aq)$

(C)  $\text{HCl}(aq)$

(D)  $\text{KOH}(aq)$



31.



$\text{CaF}_2(s)$  dissolves in water according to the equation  $\text{CaF}_2(s) \rightleftharpoons \text{Ca}^{2+}(aq) + 2\text{F}^{-}(aq)$ . The value of  $K_{sp}$  for the dissolution is  $3.5 \times 10^{-11}$ . A student measures the concentration of  $\text{Ca}^{2+}$  ions in a saturated solution of  $\text{CaF}_2$  at various pH values and uses those values to generate the graph above. Based on the data, which of the following observations about the solubility of  $\text{CaF}_2$  is most valid?



## Unit 7 Progress Check: MCQ

- (A) It does not depend on **pH** because  $[\text{Ca}^{2+}]$  does not change between **pH 4** and **pH 12**.
- (B) It does not depend on **pH** because  $K_{sp} = [\text{Ca}^{2+}][\text{F}^-]^2$ , so as  $[\text{Ca}^{2+}]$  decreases,  $[\text{F}^-]$  increases to compensate, keeping  $K_{sp}$  constant.
- (C) It is higher at a lower **pH**; there are more  $\text{H}^+$  ions in solution at low **pH**, so **HF** forms and shifts the equilibrium reaction above to the right. ✓
- (D) It is lower at a higher **pH**; there are more  $\text{H}^+$  ions in solution at high **pH**, so **HF** forms and shifts the equilibrium reaction above to the right.

32.

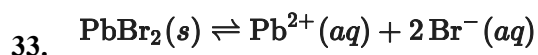
| <b>pH</b> of saturated $\text{Cu}(\text{OH})_2$ solution | Concentration of $\text{H}_3\text{O}^+$ ions | Absorbance |
|----------------------------------------------------------|----------------------------------------------|------------|
| 7.50                                                     | $3.2 \times 10^{-8} \text{ M}$               | 0.047      |
| 7.25                                                     | $5.6 \times 10^{-8} \text{ M}$               | 0.149      |
| 7.00                                                     | $1.0 \times 10^{-7} \text{ M}$               | 0.470      |

$\text{Cu}^{2+}(\text{aq})$  absorbs a certain frequency of visible light. Absorbance was measured for three saturated solutions of  $\text{Cu}(\text{OH})_2$ , each at a different **pH** at **298 K**. Based on the data recorded in the table above, which of the following conclusions about the effect of **pH** on the solubility of  $\text{Cu}(\text{OH})_2$  can be made?

- (A) When the concentration of  $\text{H}^+$  ions is increased, the solubility of  $\text{Cu}(\text{OH})_2$  increases. ✓
- (B) When the concentration of  $\text{H}^+$  ions is decreased, the solubility of  $\text{Cu}(\text{OH})_2$  increases.
- (C) The solubility of  $\text{Cu}(\text{OH})_2$  is independent of **pH**.
- (D)  $\text{Cu}(\text{OH})_2$  is soluble only at a **pH** of 7.00.



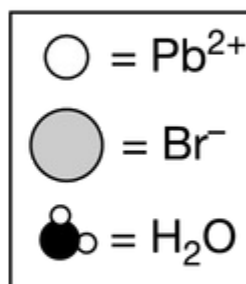
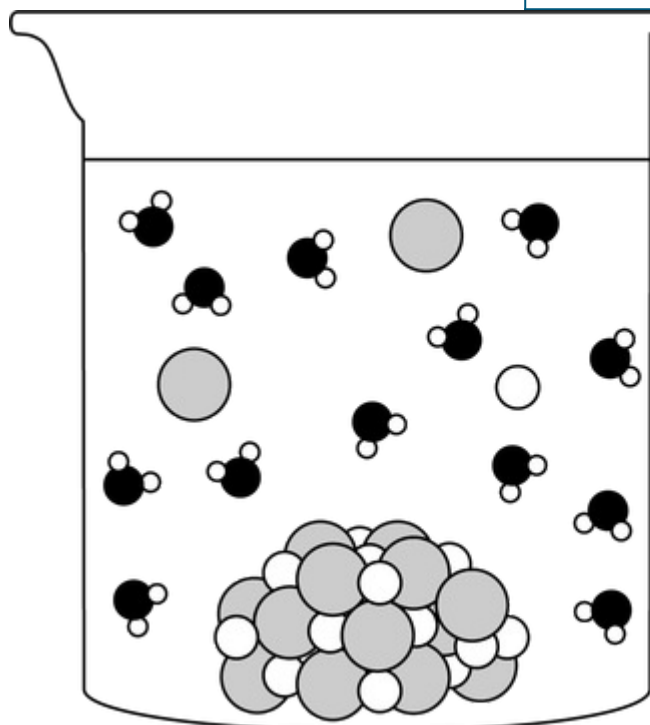
## Unit 7 Progress Check: MCQ



$$\Delta H^\circ > 0$$

$$\Delta S^\circ > 0$$

$$\Delta G^\circ > 0$$



Shown above are a

chemical equation that represents the dissolution of  $\text{PbBr}_2$  in pure water, a table of the changes in some thermodynamic properties for the process, and a particle diagram. Which of the following explains which relative change in a thermodynamic property is best illustrated by the particle diagram?

- (A) The reorganization of the water molecules around the ions illustrates that  $\Delta H^\circ > 0$  because forming strong ion-dipole interactions releases energy.
- (B) The very small amount of  $\text{Pb}^{2+}$  and  $\text{Br}^-$  ions illustrates that  $\Delta S^\circ > 0$  because entropy decreases when  $\text{PbBr}_2$  dissolves.
- (C) The very small amount of  $\text{Pb}^{2+}$  and  $\text{Br}^-$  ions illustrates that  $\Delta G^\circ > 0$  because the dissolution of  $\text{PbBr}_2$  is not a favorable process. ✓
- (D) The very large amount of solid that remains undissolved illustrates that  $\Delta G^\circ > 0$  because the dissolution of  $\text{PbBr}_2$  is a favorable process.